Hodgkin-Huxley Neuron Model Report

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1 Introduction

The Hodgkin–Huxley (HH) model is a biophysical description of the squid giant axon membrane that explains the generation and propagation of the action potential through voltage-gated sodium and potassium channels. The project consists of a computational study of a HH neuron, along with a series of experiments on gating kinetics and spiking.

2 Equations used

The membrane capacitance was calculated with the standard equation:

$$C_m \frac{dV}{dt} = I_{\text{ext}}(t) - \Big[I_{\text{Na}} + I_{\text{K}} + I_{\text{L}} \Big]$$

Where we find the ionic currents, defined by:

$$\begin{aligned} I_{\text{Na}} &= \bar{g}_{\text{Na}}\, m^3 h\, (V - E_{\text{Na}}) \\ I_{\text{K}} &= \bar{g}_{\text{K}}\, n^4\, (V - E_{\text{K}}) \\ I_{\text{L}} &= g_{\text{L}}\, (V - E_{\text{L}}) \end{aligned}$$

The gating variables follow this equation, which can be rewritten with steady states and time constants:

$$\begin{aligned} \frac{dx}{dt} &= \alpha_x(V)\,(1 - x) - \beta_x(V)\, x, \qquad\qquad x \in \{m, h, n\} \\ \frac{dx}{dt} &= \frac{x_\infty(V) - x}{\tau_x(V)} \\ x_\infty(V) &= \frac{\alpha_x(V)}{\alpha_x(V) + \beta_x(V)} \qquad\qquad \tau_x(V) = \frac{1}{\alpha_x(V) + \beta_x(V)} \end{aligned}$$

Which then gives us each activation/inactivation variable:

$$\alpha_m(V) = 0.1\, \frac{V + 40}{1 - \exp\big(-(V + 40)/10\big)}, \qquad \beta_m(V) = 4\, \exp\big(-(V + 65)/18\big),$$

$$\alpha_h(V) = 0.07\, \exp\big(-(V + 65)/20\big), \qquad \beta_h(V) = \frac{1}{1 + \exp\big(-(V + 35)/10\big)},$$

$$\alpha_n(V) = 0.01\, \frac{V + 55}{1 - \exp\big(-(V + 55)/10\big)}, \qquad \beta_n(V) = 0.125\, \exp\big(-(V + 65)/80\big).$$

The euler discretization equation is as follows:

$$V_{k+1} = V_k + \Delta t\, \frac{I_{\text{ext}}(t_k) - \big(I_{\text{Na}} + I_{\text{K}} + I_{\text{L}}\big)_k}{C_m},$$

$$x_{k+1} = x_k + \Delta t\, \frac{x_\infty(V_k) - x_k}{\tau_x(V_k)}.$$

In the temperature scaling section, we use the following equations:

$$
\begin{aligned}
k(T_{\text{exp}}) &= k(T_{\text{ref}})\; Q_{10}^{\frac{T_{\text{exp}}-T_{\text{ref}}}{10}}, \\
& \quad SQ_{10}^{\frac{T_{\text{exp}}-T_{\text{ref}}}{10}}, \\
\alpha_x^*(V) &= S\,\alpha_x(V), \qquad & \beta_x^*(V) &= S\,\beta_x(V), \qquad & x \in \{m,h,n\}, \\
x_\infty(V) &= \frac{\alpha_x(V)}{\alpha_x(V)+\beta_x(V)}, \qquad & x_\infty^*(V) &= x_\infty(V), \\
\tau_x(V) &= \frac{1}{\alpha_x(V)+\beta_x(V)}, \qquad & \tau_x^*(V) &= \frac{\tau_x(V)}{S}.
\end{aligned}
$$

3 Experiments and analysis of the model

A series of experiments were performed on the HH model in order to discern how gating kinetics and spiking work in a Hodgkin-Huxley model and their relationship.

3.1 *Current-clamp (brief pulse, long step, ramp*

Brief pulse: we tested whether a short depolarizing pulse can elicit an action potential and how spike probability depends on pulse amplitude and duration. The results showed that a brief, sufficiently strong pulse kicks off Na-driven positive feedback (m↑) to generate one AP; concurrent Na inactivation (h↓) and delayed K activation (n↑) terminate the spike and create the undershoot. Because the pulse is short and n remains elevated while h is depressed, the membrane is less excitable right after the spike, so you get exactly one spike.

Long step: we probed steady firing under a sustained depolarization to characterize excitability and basic timing (ISI / rate). The results showed that a sustained positive current shifts the balance toward inward Na current enough to cross threshold on each cycle; delayed K activation repolarizes and sets the inter-spike interval. With these classic HH parameters the cell shows tonic spiking for this input level.

Ramp: we tested the threshold accommodations (i.e.: whether a slowly rising input raises the effective threshold compared to a step). Per the results, with a slow ramp, K activation (n) rises and Na inactivation (h) falls while the voltage creeps up. This pre-conditions the membrane into a less excitable state, so the neuron fails to spike even though the final current (12 μA/cm²) is higher than the step case that produced robust spiking at 10 μA/cm². That's classic accommodation. If the ramp were faster (or the hold were longer to let h recover / n relax), a spike would likely appear at a lower effective current.

3.2 *Firing rate and DC current*

The next experiment consisted of measuring the steady firing rate vs DC current and estimating the rheobase. The finite-frequency onset confirms type II excitability for standard HH parameters: once the threshold is crossed, the system settles onto a limit cycle with non-zero frequency. The rate increases moderately with current, because stronger drive shortens the repolarization/charging phases each cycle; K activation still governs the ISI. The I= 6.0 example illustrates how transient spikes can appear below rheobase but do not count toward sustained firing once the system relaxes.

3.3 *Refractory periods*

We tested how quickly the neuron can fire again after an action potential. Two identical depolarizing pulses are delivered with an onset-to-onset gap . By determining whether a second spike occurs and how long it takes to appear, we quantify refractory recovery driven by Na inactivation (h) and K activation (n). This is relevant for temporal coding: closely spaced inputs

may fail unless the membrane has recovered sufficient excitability. The results yielded that with two identical 1 ms, 10 μA·cm² pulses, the neuron needs 15 ms to fire twice. Second-spike latency drops from 3.4 ms at the boundary to 2.1 ms by 20 ms, then plateaus.

3.4 *Strength and duration*

We measured the minimum current needed to elicit at least one spike as a function of pulse width (d). For each (d), we deliver a single square pulse (onset at 10 ms) and use bisection to find the threshold amplitude. We then fit the Weiss–Lapicque model to estimate the S–D rheobase (asymptotic current for long pulses) and the chronaxie. The S–D rheobase is defined by single-spike threshold and can differ from the DC (f–I) rheobase defined by sustained firing. The findings are that threshold current decreases with pulse width and follows:

$$I_{\mathrm{th}}(d) \approx I_{\mathrm{rh}} \left(1 + \frac{\tau}{d}\right), \qquad I_{\mathrm{rh}} \approx 1.20\ \mu\mathrm{A\,cm^{-2}}, \quad \tau \approx 5.18\ \mathrm{ms}.$$

This captures the neuron's temporal integration: short pulses need more current, while long pulses approach a low asymptotic threshold.

3.5 *Voltage clamp*

Unlike current clamp, here we prescribe the membrane voltage $V(t)$ and integrate only the gating variables m, h, n. We apply a step from a holding potential of $-65\,\mathrm{mV}$ to $0\,\mathrm{mV}$ over $t \in [10, 30]\,\mathrm{ms}$ and compute the resulting ionic currents (equations above). At $V = 0\,\mathrm{mV}$, I_{Na} is inward (negative) and transient, whereas I_{K} is outward (positive) and delayed. We repeat the step under tetrodotoxin (TTX; $g_{\mathrm{Na}} = 0$) and tetraethylammonium (TEA; $g_{\mathrm{K}} = 0$) to isolate each component. The results showed that voltage-step responses match classic HH behavior:

- Na^+: fast activation m leads to an early inward peak; inactivation h makes this transient.
- K^+: slow activation n produces a delayed outward current; minimal inactivation makes it sustained during the step with a tail upon return to rest.

Tetrodotoxin (TTX) and tetraethylammonium (TEA) remove the expected component, confirming the ionic identities and kinetics that underlie the action potential upstroke (Na) and repolarization/afterhyperpolarization (K) observed in current-clamp experiments.

3.6 *Steady state I-V curve*

We mapped the steady-state ionic I–V relation to locate the resting potential and assess stability. For each clamped voltage V, the gates are fixed at their steady state $x_\infty(V)$ and the currents are

$$\begin{aligned} I_{\mathrm{Na}} &= \bar{g}_{\mathrm{Na}}\, m_\infty^3(V) h_\infty(V)\,[V - E_{\mathrm{Na}}] \\ I_{\mathrm{K}} &= \bar{g}_{\mathrm{K}}\, n_\infty^4(V)\,[V - E_{\mathrm{K}}] \\ I_{\mathrm{L}} &= g_{\mathrm{L}}\,[V - E_{\mathrm{L}}] \\ I_{\mathrm{tot}} &= I_{\mathrm{Na}} + I_{\mathrm{K}} + I_{\mathrm{L}} \end{aligned}$$

with outward current plotted positive. The total current $I_{\mathrm{tot}}(V)$ crosses zero at $V^* \approx -65\,\mathrm{mV}$, identifying the resting potential for $I_{\mathrm{ext}} = 0$; the local slope is positive, $\frac{dI_{\mathrm{tot}}}{dV}\big|_{V^*} > 0$, indicating a stable fixed point. Near rest, $I_{\mathrm{K}} > 0$ (since $V - E_{\mathrm{K}} > 0$), $I_{\mathrm{L}} < 0$ (since $V < E_{\mathrm{L}}$), and I_{Na} is small and inward because m_∞ is tiny while h_∞ is moderate. As V is depolarized, $n_\infty(V)$ rises and I_{K} dominates, making I_{tot} strongly outward, whereas $h_\infty(V)$ decreases so I_{Na} remains small. This steady-state picture matches the voltage-clamp traces (transient Na, sustained K) and explains the return to $-65\,\mathrm{mV}$ in current clamp.

3.7 *Current clamp, pharamcology demo*

We injected the same depolarizing step (10–110 ms) under three conditions: Control, TTX ($g_{\mathrm{Na}} = 0$) blocking fast Na^+, and TEA ($g_{\mathrm{K}} = 0$) blocking delayed rectifier K^+. This reveals how each channel family shapes spiking under current clamp.

- Control. The step drives tonic spiking: repeated action potentials throughout the stimulus window. This reflects the fast positive feedback of Na activation (m) to generate the upstroke and the slower K activation (n) that repolarizes and sets the inter–spike interval.
- TTX ($g_{\mathrm{Na}} = 0$). No action potentials occur. Only a small, passive-like depolarization at onset/offset is visible. Removing the inward Na current eliminates the upstroke, so the neuron cannot reach threshold under the same current.
- TEA ($g_{\mathrm{K}} = 0$). A large initial spike appears, followed by a sustained depolarized plateau and no further spikes (depolarization block). Without the outward K current, repolarization fails; Na inactivation (h) then accumulates and prevents repetitive firing until the step ends.

Overall, Na current is necessary for spike initiation (upstroke), whereas K current is necessary for repolarization and for sustaining repetitive spiking under steady depolarization.

3.8 *Temperature scaling*

To model temperature effects we apply a single Q_{10} factor to all gating rates (not to reversal potentials or maximal conductances). Because both forward and backward rates are scaled equally between the experimental temperature T_{exp} and a reference T_{ref}, the steady-state gates $x_\infty(V)$ remain unchanged, while the time constants $\tau_x(V)$ decrease by the same factor. In practice, gates chase their targets faster at higher temperature. Our findings show that:

- Faster activation/inactivation: τ_m, τ_h, and τ_n are uniformly shorter.
- Spike shape: narrower action potentials and shorter latency to first spike.
- Firing: higher steady firing rates for a given input current; shorter absolute/relative refractory periods.
- Subthreshold: resting potential and the $x_\infty(V)$ curves are essentially unaffected by rate-only scaling.

3.9 *Phase plane with nullclines*

Assuming the fast gates are slaved to voltage, $m \approx m_\infty(V)$ and $h \approx h_\infty(V)$, the Hodgkin–Huxley dynamics can be visualized in the (V, n) plane. We plot the two nullclines:

- $dV/dt = 0$: points where the net ionic current balances the external current (here $I_{\mathrm{ext}} = 0$); this yields a curve $n = n_V(V)$.
- $dn/dt = 0$: the slow-gate nullcline $n = n_\infty(V)$.

Their intersection gives the fixed point (rest). We then overlay the trajectory $(V(t), n(t))$ produced by a brief current pulse (10–11 ms) to show the spike excursion and recovery. Under Q_{10} temperature scaling the nullclines do not shift; only the traversal speed along the trajectory changes because the time constants are rescaled. The nullclines locate a stable rest; the pulse propels the system past the V-nullcline knee, producing a clockwise spike excursion (right $\rightarrow$ up $\rightarrow$ left $\rightarrow$ down) that returns to rest as the slow variable n relaxes — illustrating fast V vs slow n dynamics of the Hodgkin-Huxley neuron.

4 Conclusion

We implemented and validated a classic single-compartment Hodgkin-Huxley (HH) neuron and used a set of targeted experiments to connect biophysical mechanisms to observable dynamics. Voltage clamp and steady-state I-V confirmed the expected separation of roles: a fast, transient inward Na^+ current and a delayed, sustained outward K^+ current; the I-V zero crossing predicted a stable resting potential near $-65\ \mathrm{mV}$. Under current clamp, a long step produced tonic spiking with Type-II onset; the f-I curve jumped from 0 to a finite frequency just above the repetitive-firing rheobase ($I_{\mathrm{rh,step}} \approx 6.25\ \mu\mathrm{A\,cm^{-2}}$), with rates $\sim$ 65–73 Hz at 9–12 $\mu\mathrm{A\,cm^{-2}}$.

Short-pulse tests quantified temporal excitability. The strength-duration relation followed the Weiss-Lapicque law with single-spike rheobase $I_{\mathrm{rh}} \approx 1.20\ \mu\mathrm{A\,cm^{-2}}$ and chronaxie $\tau \approx 5.18\ \mathrm{ms}$, showing that brief inputs require disproportionately larger currents. Paired pulses revealed refractory recovery: a second spike at fixed amplitude appeared only for onset gaps $\Delta \gtrsim 15\ \mathrm{ms}$, with latency decreasing as Δ grew. Ramp stimuli demonstrated accommodation: a slow 0$\rightarrow$12 $\mu\mathrm{A\,cm^{-2}}$ ramp failed to spike, even though a 10 $\mu\mathrm{A\,cm^{-2}}$ step fired robustly, highlighting the impact of input timescale on threshold.

Temperature (Q_{10}) scaling left $x_\infty(V)$ unchanged but shortened all gating time constants, producing earlier, narrower, lower-amplitude spikes and higher firing rates at the same current. A phase-portrait view in the (V, n) plane tied these behaviors to geometry: the fixed point at the nullcline intersection and a spike excursion that crosses the V-nullcline knee (a practical threshold) and returns as n relaxes.

Limitations and outlook. The model is deterministic and single-compartment with uniform Q_{10} scaling and constant conductances; channel noise, morphology, synaptic conductances, and adaptation currents are omitted.